

The normal-crossing example with Gaussian data

Claude, with Daniel Murfet

2026-09-07

Abstract

Unit 219 completes the H2-lite example of Astra #24 on the probabilistic side. Unit 218 gave the posterior moment generating function limit $\mathbb{E}_{\text{post}}[e^{\theta\sqrt{n}x_0x_1}] \rightarrow e^{z\theta+\theta^2/2}$ along data sequences whose empirical phase $Z_n = n^{-1/2} \sum_i y_i$ converges to z . Here the data are random, Y_i i.i.d. $N(0, 1)$, so Z_n has mean 0 and variance 1 for every n and Chebyshev gives $P(|Z_n| \geq M) \leq M^{-2}$. Together with *uniform* convergence of the posterior moment generating function on compact phase sets (eventual equi-Lipschitz dependence on the phase from unit 209, pointwise limits from unit 218, finite net; a quotient lemma for uniformly convergent numerators over denominators bounded below) this gives

$$\mathbb{E}_{\text{post}}[e^{\theta\sqrt{n}x_0x_1}] - e^{Z_n\theta+\theta^2/2} \longrightarrow 0 \quad \text{in } P\text{-probability}$$

(ncFluctMGF_tendstoInMeasure): the posterior law of $\sqrt{n}x_0x_1$ is asymptotically $N(Z_n, 1)$ with a random standard-normal centre. The only probabilistic inputs are the second moment of the data and pairwise independence (variance of the sum). Zero `sorry/axiom`; 9096 jobs.

1 The things formalised

```
theorem tendstoUniformlyOn_of_lipschitz_eventually : compact K,
  eventual equi-Lipschitz F n, continuous f, pointwise F n x → f x on K
  TendstoUniformlyOn F f 1 K
theorem tendstoUniformlyOn_div_of_pos : F → f, G → g uniformly, |f| ≤ B, c ≤ g
  F/G → f/g uniformly
def ncNorm N z := ( _{symBox 2} K_{N,z} ) / leadScale ... N
def ncLimit z := 2 (0)√(2) e^{z^2/2}
def ncRatio N z := ( K_{N,z+} ) / ( K_{N,z} )
theorem ncFluctMGF_eq : ncFluctMGF n y = ncRatio √n (ncPhase n y)
theorem ncNorm_lipschitz_eventually : L, n, z z' ∈ [-M,M],
  |ncNorm ... z - ncNorm ... z'| ≤ L|z-z'|
theorem tendstoUniformlyOn_ncNorm / tendstoUniformlyOn_ncRatio (on Icc (-M) M)
def ncPhaseRV Y n := ncPhase n (fun i : Fin n => Y i) -- (√n)^{-1} • i, Y i
theorem variance_ncPhaseRV (hY : i, HasLaw (Y i) (gaussianReal 0 1) P)
  (hind : iIndepFun Y P) : Var[ncPhaseRV Y n; P] = 1
theorem meas_abs_ncPhaseRV_ge_le :
  P { | M | ncPhaseRV Y n | } ENNReal.ofReal (1 / M ^ 2)
theorem ncFluctMGF_tendstoInMeasure : TendstoInMeasure P (fun n =>
  ncFluctMGF n (fun i => Y i) - exp (ncPhaseRV Y n * + ^2/2)) atTop 0
```

2 Mathematics

Uniformity in the phase is the only new analytic point: on $[-M, M]$ the four quadrant integrals are eventually $2L_\sigma$ -Lipschitz in z (phase $\varepsilon_\sigma \cdot 2z$, $|\varepsilon_\sigma| = 1$), so the normalised denominator is eventually

equi-Lipschitz, and equi-Lipschitz plus pointwise convergence to a continuous limit on a compact set is uniform convergence (finite δ -net). The numerator is the denominator at $z + \theta$, uniform on $[-M - |\theta|, M + |\theta|]$; the limiting denominator is bounded below by $2\rho(0)\sqrt{2\pi} > 0$, so the quotient converges uniformly. Then on $\{|Z_n| < M\}$ the error is eventually $< \varepsilon$, and $P(|Z_n| \geq M) \leq M^{-2}$ is made smaller than any δ by choosing $M = \max(1, 1/\delta)$.

3 Lean notes

- `TendstoInMeasure` is now stated with `edist` and $\varepsilon \in [0, \infty]$; use `tendstoInMeasure_iff_norm` to work with real ε and $\|f - g\|$.
- Sums of random variables indexed by `Fin n` with `Y :  $\Omega \rightarrow$  : pass (f := fun i : Fin n => Y i) / (X := ...)` to `memLp_finsetSum'`, `integral_finsetSum`, `IndepFun.variance_sum`, else the index type unifies to `.`. Pairwise independence from `hind.indepFun (Fin.val_injective.ne hij)`.
- `MemLp` from a law: `memLp_id_gaussianReal 2` transported through `HasLaw.map_eq` and `memLp_map_measure_iff astronglyMeasurable_id hY.aemeasurable`.
- `omit [IsProbabilityMeasure P] in` must precede the docstring, not sit between the docstring and the theorem (parse error). A lake `build | grep` pipeline returns `grep`'s status: the broken file was committed once (13c8440) before the gated rebuild caught it (fixed in 9e94412).

4 Status

H2-lite complete at the in-probability level (Headline XX, XX'). Possible last step: the exact law $Z_n \sim N(0,1)$ via characteristic functions (`iIndepFun.charFun_map_sum_eq_prod`, `charFun_gaussianReal`) to state convergence in distribution to $e^{Z\theta + \theta^2/2}$, $Z \sim N(0,1)$; then freeze per Astra #24.